SCHOOL OF INFORMATION TECHNOLOGY

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A SYNOPSIS OF MAJOR PROJECT- II
ON

AI-Powered Video Knowledge Mapper

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Fulfillment Of the Requirements for The Award of The Degree Of

BACHELOR OF TECHNOLOGY

Computer Science Engineering (With Specialization In

Artificial Intelligence And Machine Learning)

By

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Introduction

The “AI Powered Video Knowledge Mapper” is an advanced generative AI project designed to convert long-form educational videos into structured visual knowledge representations. The system automatically analyzes video content and generates mind maps (concept flowcharts) that highlight key ideas and relationships within the video. By leveraging modern large language models (LLMs) and natural language processing (NLP), the platform transforms unstructured video transcripts into organized conceptual diagrams that improve comprehension and knowledge retention.

The system integrates transcript extraction, semantic understanding, and structured output generation into a unified workflow. Video transcripts are extracted using automated caption retrieval tools and processed using lightweight generative AI models. The AI model interprets the transcript and generates structured mind map data in JSON format, representing nodes (concepts) and edges (relationships). These structured outputs are then rendered visually using interactive frontend technologies that allow dynamic exploration of the generated knowledge.

A key feature of the project is the use of adjustable generation parameters such as temperature, enabling users to control the level of abstraction and diversity in generated mind maps. This makes the system adaptable to different learning preferences, ranging from concise summaries to more exploratory concept maps.

Beyond its technical implementation, the project promotes hands-on learning in modern AI engineering, including API integration, structured output generation, frontend visualization, and full-stack development. By bridging theoretical AI concepts with practical implementation, the project provides a strong foundation for future work in AI-powered learning systems, intelligent summarization, and knowledge visualization tools.

Rationale

With the rapid growth of online educational platforms, video-based learning has become a dominant medium for knowledge acquisition. However, extracting meaningful insights from long videos remains a time-consuming and cognitively demanding task. Students often struggle to identify key concepts, relationships, and hierarchical structures within lengthy video content.

The "AI Powered Video Knowledge Mapper" addresses this challenge by leveraging generative AI to automatically transform video transcripts into structured visual mind maps. Instead of manually summarizing videos, users receive instantly generated conceptual diagrams that present information in a structured and easily understandable format.

Objectives

- **Develop an AI-Based Knowledge Mapping System:** Build a platform capable of extracting video transcripts and converting them into structured mind maps automatically.
- **Implement Generative AI Integration:** Utilize modern generative AI APIs to interpret transcripts and generate hierarchical concept structures in structured formats.
- **Enable Interactive Visualization:** Create an intuitive frontend that displays generated mind maps as interactive node-edge diagrams.
- **Enhance Technical and Analytical Skills:** Provide hands-on experience in full-stack AI development, structured output design, and frontend visualization technologies.

Feasibility Study

- **Technical Feasibility**

  The AI Powered Video Knowledge Mapper is technically feasible due to the availability of mature web frameworks, transcript extraction tools, and scalable generative AI APIs. The backend can be built using Python with FastAPI for efficient API routing and integration with transcript and AI services. Video captions can be extracted reliably using transcript APIs, while lightweight generative AI models process transcripts without requiring local training. Structured JSON outputs enable seamless frontend visualization using React-based libraries, ensuring a modular and easily implementable architecture within academic timelines.

- **Need for the Project**

  The rapid growth of online educational videos has created a strong need for tools that can extract structured knowledge efficiently. Learners often find it difficult to identify key concepts and relationships from long-form videos, leading to reduced comprehension and retention. This project addresses the gap by automatically transforming transcripts into structured visual mind maps that highlight conceptual hierarchies and connections. By combining generative AI with interactive visualization, the system improves knowledge accessibility and supports modern AI-assisted learning approaches, making it both academically relevant and practically valuable.

- **Significance**

  The project is economically viable due to its reliance on open-source technologies and lightweight infrastructure. Frontend and backend frameworks such as React.js and FastAPI are freely available, reducing development costs. The system uses cloud-based generative AI APIs instead of locally trained models, eliminating the need for high-end GPUs or expensive hardware. Development and testing can be performed on standard academic laptops, while structured JSON outputs and efficient API usage help control operational costs. This ensures the project remains affordable, scalable, and suitable for student-level implementation.

Methodology / Planning of work

❖ Research Approach

The project follows an applied research methodology focused on building a real-world generative AI application. It emphasizes practical experimentation with transcript processing, prompt engineering, structured output generation, and interactive visualization. The primary goal is to design a modular and scalable system capable of transforming video content into meaningful visual knowledge maps.

❖ Project Units

The development process will be divided into following functional units:

1. **Transcript Extraction Unit:** Responsible for retrieving captions from YouTube videos using transcript APIs and preprocessing text for AI consumption.

2. **Generative AI Processing Unit:** Handles prompt design, API integration, and conversion of transcripts into structured JSON mind map representations.

3. **Backend & Caching Unit:** Implements FastAPI-based services to manage data flow between the frontend, transcript services, and AI APIs, while utilizing a PostgreSQL database to cache generated knowledge graphs for optimized performance and reduced API credit costs.

4. **Visualization Unit:** Focuses on rendering structured JSON outputs into interactive mind maps using React Flow and automatic layout engines like Dagre.

❖ Steps to Achieve Project Objectives

1. **Transcript Processing:** Implement transcript extraction and preprocessing pipelines using caption APIs.

2. **AI Processing & Storage:** Integrate generative AI APIs to output structured JSON representations of nodes and relationships, and store these outputs in the database to prevent redundant API calls for previously processed videos.

3. **Frontend Development:** Develop a React-based interface that accepts video URLs and displays interactive mind maps using React Flow.

4. **Testing and Refinement:** Evaluate output accuracy, refine prompts, optimize API usage, and improve user experience.

Software Required

Category	Tools / Technologies
Frontend Framework	React.js
Visualization Library	React Flow
Layout Engine	Dagre
Backend Framework	FastAPI
Transcript Extraction	YouTube Transcript API
Generative AI Model	Google Gemini API
Database	PostgreSQL
Development Tools	VS Code / Jupyter Notebook
Version Control	Git / Github

Expected Outcomes

1. **AI-Powered Knowledge Mapping System:**
   A fully functional application capable of transforming educational videos into structured, interactive mind maps automatically.

2. **Functional Automation:**
   Practical exposure to working with LLM APIs, structured JSON outputs, and real-world AI system integration.

3. **Skill Development:**
   Enhanced proficiency in prompt engineering, backend architecture, and interactive visualization technologies, preparing students for careers in AI-powered software development and intelligent learning systems.

Signatures Of Students

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